Veritrace

Evidence-first fact-checking for Indian public claims

Theme 1: Civic Pulse, Digital Governance & Public Information

Team: add names here

PROBLEM

False claims spread faster than checks

1

Viral claims

Forwards about free schemes or new taxes reach people before any correction.

2

Scattered facts

Official information exists, but it is spread across many sites and hard to read.

3

Risky chatbots

AI chatbots can sound certain while being wrong, and rarely show exact proof.

Who it affects

Citizens, benefit applicants and small businesses need facts they can trust. Fact-checkers, newsrooms and public-information teams can't check every claim by hand.

SOLUTION AND KEY FEATURES

Three ways to use Veritrace

1

Paste a claim

On the website. Veritrace checks up to 3 claims and skips opinions.

2

Select text on any page

In Chrome, right-click Verify with Veritrace. The result opens in a side panel.

3

Ask a question

On the website, ask about schemes, rights and rules, with cited sources.

Each claim gets 1 of 6 results

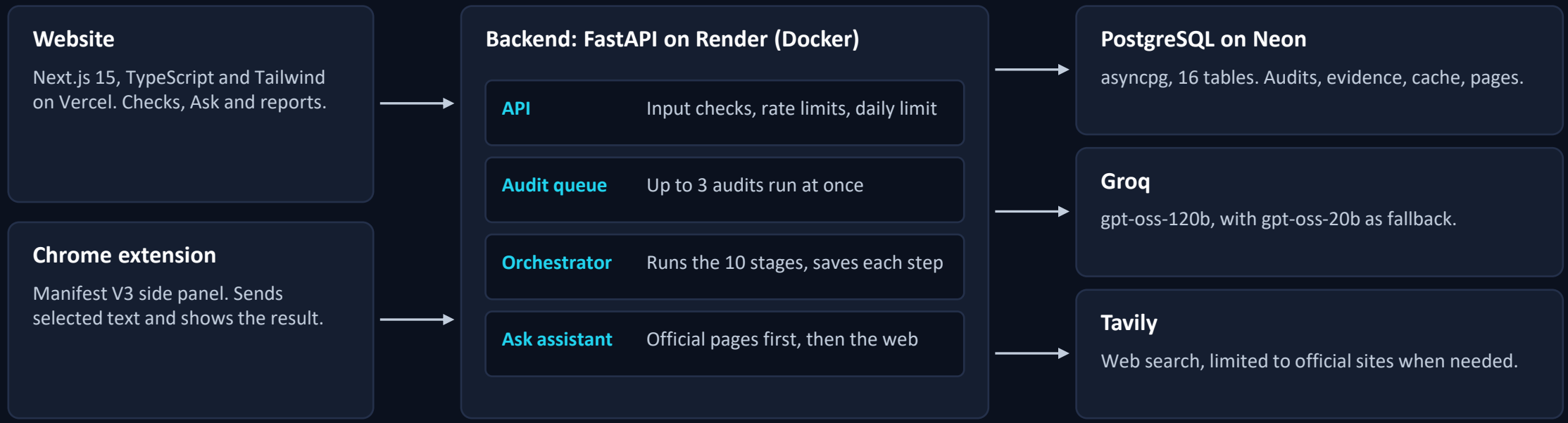
Confirmed	the claim is correct
Incorrect	the evidence contradicts it
Misleading	true, but missing context
Sources disagree	strong sources conflict
Unconfirmed	not enough evidence
Not checked	the system could not finish

Each comes with a summary, exact quotes, source links and a confidence level.

Also: live progress, Cancel, an estimated time left, and a list of past audits.

TECHNICAL ARCHITECTURE

One backend serves the website and the extension



Caching

Search results are kept, except for recent claims. Repeat reports are reused for 7 days. Page text is compressed.

Limits

12,000 characters per input, 18 search attempts per audit, and 30 requests an hour per address.

Safety

URLs are checked before fetching. At most 3 redirects, 5 MB and 50 PDF pages per document.

Why this architecture works

One pipeline
The website and extension run the same checks through one API.

Traceable
AI output passes through code, and each verdict records its rule.

Fault tolerant
Each step is saved. If one claim fails, the others still finish.

Low cost
Repeat checks reuse saved results, with no new search or AI calls.

TECHNICAL PIPELINE

How each claim is checked, in 10 stages

1 Extract claims
AI

Up to 3 factual claims, with key details, numbers and dates.

2 Classify
AI

Official record, scientific or reported event, plus topic.

3 Plan searches
AI + template

4 queries per claim, shown below.

4 Search
Tavily

Run in parallel. 8 results each, 20-second limit.

5 Fetch and read
Python

Trafilatura for pages, PyMuPDF for PDFs. 5 documents per claim.

6 Rank passages
BM25

4-sentence windows. Top 8, at most 2 per document.

7 Label evidence
AI

Marks numbered sentences as supporting or contradicting.

8 Validate
Python

Rebuilds quotes, then checks names, dates, scope and numbers.

9 Fill gaps
AI + Python

If details are missing, one round of up to 2 extra searches.

10 Decide and report
Rules + AI

Rules set the verdict. AI writes the summary.

The 4 searches look for evidence for and against each claim

For the claim, official sites

The claim's own words, limited to government and listed official sites.

For the claim, open web

An AI-written query for direct proof.

Against the claim, open web

An AI-written query for denials, corrections and exceptions.

Against the claim, official sites

The claim plus "clarification correction".

Social media sites are blocked from open-web searches.

VERIFICATION AND VERDICT ENGINE

Veritrace validates every quote, then rules set the verdict

Each evidence item is checked

- Quote is rebuilt from saved sentence IDs
- The same person, body or scheme is named
- Location and year match the claim
- "Only" or "eligible" limits an "all" claim
- "May" or "proposed" can't confirm a fact
- Numbers match after units: lakh, crore, ₹, %

Valid counts toward the verdict.
Contextual is shown, but can't prove a claim.
Invalid is rejected.

Rule	Condition	Claim result (6)
R0	The analysis failed	**Not checked**
R1	Strong evidence on both sides	**Sources disagree**
R2	Strong evidence against	**Incorrect**
R3	Official source, different period or group	**Misleading**
R4	Strong support, every key detail answered	**Confirmed**
R5	None of the above	**Unconfirmed**

Strong evidence: 1 official source, or 2 distinct publishers. Social media never counts.

High confidence: an official source, no gaps, exact numbers and no dispute.

Tie-break: official support outranks a lone unofficial contradiction.

Overall result (8): for the full text, the 5 findings above, or Partly confirmed, Incomplete or Nothing to check.

RESULTS AND FUTURE SCOPE

Tested and live today, with a clear next step

Results

6 of 6

test claims reached the expected result

315

automated tests passing

A small sample of known answers chosen by us, not a general accuracy figure. Checks took 16 to 133 seconds on live sources, with no mock data.

Future scope

- Hindi and regional languages
- Screenshots and image history
- Ranked feed of spreading claims
- Knowledge graph for entities and sources
- Media signals as one input to deepfake checks

Try it: veritrace-iota.vercel.app

Veritrace doesn't detect deepfakes yet. We built the verification core first, because evidence you can trust matters most.